PALATIAL POOL SERVICES
Proprietary Technology

Wi-Fi Radar

Invisible Safety Monitoring Through RF Sensing

Wi-Fi Radar transforms standard wireless signals into an invisible safety monitoring network around your pool. Without any cameras or optical surveillance, the system detects human presence, movement patterns, and even micro-movements like breathing — providing continuous occupancy monitoring, unauthorised access alerts, and structural integrity tracking, all while preserving complete privacy.

Key Benefits

- Continuous pool area monitoring without any cameras
- Detects presence through walls, fences, and pool covers
- Unauthorised access alerts sent instantly to your phone
- Micro-movement detection for advanced safety scenarios
- Long-term structural monitoring of pool surroundings

At a Glance

Technology: Channel State Information (CSI) analysis across 2.4/5/6 GHz

Hardware: Custom ESP32-S3 nodes with enterprise access points

Detection: Occupancy, movement patterns, micro-movements, structural shifts

Processing: Butterworth filters, PCA, Doppler shift analysis

Alerting: Real-time via MQTT over TLS to Comms Hub

Privacy: No optical data — pure RF signal analysis, anonymised processing

Ready to upgrade your pool service experience?

Call **0452 157 694** or visit **beta.palatialpools.com.au**

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